Project folder anatomy

A SWAT+ project is one folder of plain-text files, usually called `TxtInOut`. The model reads every input from that folder and writes its outputs back into it. Weather data files can be kept in separate folders named in file.cio; everything else is in the one folder.

The master file

file.cio is the index of the project. Each line names a group of inputs and the files to read for it, in a fixed order. `null` means that the file is not used. Read this file first when you open a project. The full description is on file.cio.

file.cio of the `Ames_sub1` reference dataset at 62.0.1 (spacing shortened):

```
file.cio: AMES
simulation      time.sim  print.prt  null  object.cnt  null
basin           codes.bsn  parameters.bsn  carbon.bsn
climate         weather-sta.cli  weather-wgn.cli  null  pcp.cli  tmp.cli  null  null  null  null
connect         hru.con  null  null  null  null  null  null  null  null  null  null  null  null
channel         null  null  null  null  null  null  null  null
reservoir       null  null  null  null  null  null  null  null
routing_unit    null  null  null  null
hru             hru-data.hru  null
exco            null  null  null  null  null  null
recall          null
dr              null  null  null  null  null  null
aquifer         nbull  null
herd            null  null  null
water_rights    null  null  null
link            null  null
hydrology       hydrology.hyd  topography.hyd  null
structural      tiledrain.str  septic.str  filterstrip.str  grassedww.str  bmpuser.str
hru_parm_db     plants.plt  fertilizer.frt  tillage.til  pesticide.pes  null  null  null  urban.urb  septi
ops             harv.ops  graze.ops  irr.ops  chem_app.ops  fire.ops  sweep.ops
lum             landuse.lum  management.sch  cntable.lum  cons_practice.lum  ovn_table.lum
chg             null  null  null  null  null  null  null  null  null
init            plant.ini  soil_plant.ini  null  null  null  null  null  null  null  null  null  null
soils           soils.sol  nutrients.sol  null
decision_table  lum.dtl  null  null  null
regions         ls_unit.ele  null  null  null  null  null  null  null  aqu_catunit.ele  null  null  null
pcp_path        null
tmp_path        null
slr_path        null
hmd_path        null
wnd_path        null
```

The `basin` line names three files from revision 62 on (carbon.bsn was added). A file.cio written for an earlier revision has two names there and is misread by 62.0.1; rewrite it with SWAT+ Editor 4 or add the third name. The same applies to the `water_rights` line, which names the water allocation files. See file.cio for the details.

Input groups

Simulation control

File	Content
time.sim	start and end dates, time step
print.prt	which output files to write, and at which time steps
object.cnt	number of objects of each type (HRUs, channels, aquifers, …)
object.prt	extra output for individual objects (optional)

Basin

File	Content
codes.bsn	method switches (PET method, runoff method, routing, carbon model, gwflow, …)
parameters.bsn	basin-wide parameters
carbon.bsn, carbon_lyr.bsn	carbon model parameters; carbon_lyr.bsn is found from the name of carbon.bsn and is not listed in file.cio

Climate

File	Content
weather-sta.cli	weather stations and the data files each one uses
weather-wgn.cli	weather generator statistics, used to fill missing data or generate whole records
pcp.cli, tmp.cli, slr.cli, hmd.cli, wnd.cli	lists of the data files for each variable
*.pcp, *.tmp, *.slr, *.hmd, *.wnd	daily (or sub-daily) records

Objects and connections

File	Content
hru.con, rout_unit.con, aquifer.con, chandeg.con, reservoir.con, recall.con, …	position of each object and where its outflow goes
hru-data.hru	for each HRU, the names of its topography, hydrology, soil, land use, initial conditions, snow and field records
rout_unit.def, rout_unit.ele, ls_unit.def, ls_unit.ele, aqu_catunit.ele	which HRUs belong to each routing unit, landscape unit and aquifer catchment

Hydrology and topography

File	Content
hydrology.hyd	HRU hydrology parameters (lateral flow, evaporation, percolation, …)
topography.hyd	slope, slope length and related values
field.fld	field length, width and orientation

Soils

File	Content
soils.sol	soil layers and their physical properties
nutrients.sol	initial soil nutrient contents

Land use and management

File	Content
landuse.lum	land uses, each naming its plant community, management schedule, curve number set, conservation practice and structural practices
management.sch	management schedules: dated operations and decision tables
cntable.lum, cons_practice.lum, ovn_table.lum	curve numbers, USLE support practice factors, overland Manning's n
lum.dtl	decision tables for land use and management
harv.ops, graze.ops, irr.ops, chem_app.ops, fire.ops, sweep.ops	parameters of each type of operation

Databases

File	Content
plants.plt	plant growth parameters
fertilizer.frt, manure_db.frt, manure_om.frt	fertilizers and manures
tillage.til, pesticide.pes, urban.urb, septic.sep, snow.sno	tillage, pesticide, urban, septic system and snow parameters

Initial conditions

File	Content
plant.ini	plant communities at the start of the run
soil_plant.ini	which soil nutrient and constituent records each HRU starts with

Channels, aquifers, reservoirs and wetlands

File	Content
channel-lte.cha, hyd-sed-lte.cha, nutrients.cha, initial.cha	channels (the SWAT-DEG channel model, connected through chandeg.con)
aquifer.aqu, initial.aqu	aquifers
reservoir.res, hydrology.res, sediment.res, nutrients.res, initial.res, weir.res	reservoirs
wetland.wet, hydrology.wet	wetlands

Calibration

File	Content
cal_parms.cal, calibration.cal	parameter changes applied at start-up (Hard calibration)
codes.sft and the other *.sft files	Soft calibration

Optional groups (structural practices, water allocation, gwflow, salts and constituents, pesticides and pathogens, export coefficients, recall) are described in the Input Reference.

Outputs

Output files end in `_day`, `_mon`, `_yr` or `_aa` (average annual) and `.txt`, or `.csv` when CSV output is switched on in print.prt. The first part of the name gives the object (`basin_`, `lsunit_`, `hru_`, `channel_sd_`, `aquifer_`, `reservoir_`, …) and the next part the variables:

Code	Variables
`wb`	water balance
`nb`	nutrient balance
`ls`	sediment and nutrient losses
`pw`	plant growth and stress
`aqu`, `sd_cha`, `res`	aquifer, channel, reservoir

The Output Reference describes each file.

Log and check files

File	Content
`simulation.out`	run progress and the completion message
`diagnostics.out`	names that could not be matched, missing files, unexpected values
`checker.out`	summary of the inputs for each HRU, written before the simulation
`files_out.out`	list of the output files written
`area_calc.out`	area summary
`erosion.out`	header lines only in 62.0.1
`success.fin`	written only when the run completes

Check `simulation.out` and `diagnostics.out` after every first run of a new or changed project.

Sources

- File groups and default names: the `input_*` types in `src/input_file_module.f90`; `file.cio` is read by `src/readcio_read.f90`.
- Example: `refdata/Ames_sub1/file.cio`. The `aquifer` line contains `nbull`, a typing error in the shipped file; the dataset has no aquifers, so the name is never used.